

READYKIT · REQUIREMENTS

ReadyKit requirements

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/readykit		
Source	docs/03-requirements.md @ 5a3ae53		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	Fifteen measurable requirements for the tool with targets, current status and verification
0.1	2026-09-25	Amish Chadha	Initial scaffold

ReadyKit requirements

ReadyKit meets its speed, cost and licensing targets today, measured on one sample repository. It does not yet meet the targets for check coverage, Windows support, configurable branding, one-command upgrades or a published getting-started guide (Table 2). Those gaps define the TRL 3 work.

Measured values come from running kit 1.3.1 on a copy of the TremorTrace repository (three controlled documents, six-part massing model) in a 2-core Linux container with Python 3.11 on 2026-09-25. Everything else is an estimate and is marked as one.

Table 1. Requirements.

ID	Requirement	Target	Status at TRL 2	Verification
R1	Setup time from an empty folder to a first passing check	15 min or less on Linux or macOS with Python installed	Not verified; estimated 10 to 20 min, dominated by installing WeasyPrint system libraries	Timed trial by a new user
R2	Platforms	Linux, macOS and Windows (native or WSL2)	Linux met (CI on Ubuntu); macOS expected; Windows native not met (WeasyPrint needs Pango, not tested)	CI matrix on three operating systems
R3	Check time per repository	2 s or less	Met: about 0.1 s (measured)	Timed run in CI
R4	PDF render time	10 s or less for three documents	Met: about 1.6 s (measured)	Timed run
R5	Concept media render time	60 s or less for a model of up to 20 parts, no GPU	Met for 6 to 7 parts: about 5 to 7 s (measured); 20 parts not tested	Timed run on a 20-part model
R6	Check coverage	90% or more of the machine-checkable rules in the standard are checked automatically	Not met: 12 of 20 rules, 60% (Table 3)	Rule-by-rule audit against STANDARDS.md
R7	False failures	No check fails a document that follows the standard	Not verified; no false failure seen in the portfolio so far	Test corpus of conforming documents
R8	Reproducible output	Same inputs give the same PDF text and images, apart from date and commit	Not verified	Render twice and compare
R9	Offline use	Check and render run with no network	Met, except <code>viewer.html</code> , which loads its viewer script from a CDN	Run with network disabled
R10	Footprint	Vendored kit folder 2 MB or less	Met: about 1.1 MB, mostly fonts (measured)	<code>du</code> in CI
R11	Configurable identity	Organization name, website, repository owner, author and colors set in one config file, none hard-coded	Not met: footer site, repository owner and author default are hard-coded	Render a repo with a second identity
R12	Upgrade path	One command updates the kit in a repo to a new version and shows the diff	Not met: kit is copied by hand into each repo	Upgrade trial across 5 repos
R13	Agent guardrails	Check flags TRL above the cap, work beyond the cap, and decisions recorded as made without an owner	Partly met: cap and beyond-cap work checked; decision wording not checked (manual grep today)	Seeded test documents
R14	Cost to user	\$0; runs on free CI for public repositories	Met: all dependencies free and open source	Dependency license audit

ID	Requirement	Target	Status at TRL 2	Verification
R15	Documentation	Getting-started guide of 5 pages or fewer, plus a reference for every check	Not met: only STANDARDS.md and the kit README exist	Guide review by a new user

Table 2. Requirements not met at TRL 2.

ID	Gap	Proposed route (awaiting Amish)
R2	Windows native untested	Add a Windows CI job; document WSL2 as the fallback
R6	60% check coverage	Add the eight missing checks in Table 3
R11	Hard-coded identity	Move identity into a <code>readykit.yaml</code> file with the portfolio values as defaults
R12	Manual kit sync	Package the kit for pip with a <code>readykit upgrade</code> command
R13	Decision wording unchecked	Promote the portfolio's decision-wording grep into the checker
R15	No getting-started guide	Write it at TRL 3

Table 3. Machine-checkable rules in STANDARDS.md and whether the checker covers them today.

Rule	Checked
Required front matter fields present	Yes
Document ID matches PRJ-TYP-NNN	Yes
Document ID unique within the repository	No
Version format (quoted, n.n)	Yes
Version equals newest revision row	Yes
Every revision row complete	Yes
Status in the allowed set	Yes
Released documents at 1.0 or later	Yes
No em dash anywhere in the repository	Partly (controlled documents only), counted as No
TRL evidence present for the claimed level	Yes
TRL and target within the phase cap	Yes
Required concept media present	Yes (warning at TRL 2)
Concept media carry the "not for fabrication" label	No
BOM columns and numbering match the exploded view	No
BOM fully priced at TRL 3	Yes
Evidence files listed in project.yaml exist	Yes
Figures and tables numbered and captioned	No
Safety note present where hazard keywords appear	No
README sections in the required order	No
SI unit formatting (space between value and unit)	No